Trust Architecture:

A Constitutional Framework for AI Infrastructure

Version 3

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Preamble

Artificial intelligence is becoming integrated into the systems, institutions, and infrastructure that shape modern life. As this integration deepens, questions of capability are no longer sufficient. Equally important are the conditions under which those capabilities operate — and the degree to which people can trust, participate in, understand, influence, and benefit from the systems they increasingly depend upon.

Throughout history, major advances in human capability have been accompanied, often belatedly, by new social, institutional, and constitutional arrangements. The challenge is to ensure that capable systems develop alongside the conditions that support trust, legitimacy, accountability, participation, dignity, privacy, stewardship, and continuity.

This paper explores Trust Architecture as a constitutional framework for AI infrastructure. It identifies the conditions that support genuine trust and legitimate participation within increasingly intelligent and interconnected systems, and contributes to the broader conversation about the constitutional foundations upon which trustworthy infrastructure can be built, maintained, corrected, and renewed.

The framework recognizes that its principles will be adopted unevenly and applied imperfectly. Disagreement, conflict, misuse, and competing interests are inevitable. Constitutional frameworks provide structure through which trust, participation, accountability, and legitimacy can be more clearly understood, evaluated, and advanced.

The framework that follows is presented as an architectural perspective. Its constitutional conditions illuminate how trust becomes part of the infrastructure itself — rather than an expectation placed upon it after the fact.

Introduction

Artificial intelligence is moving beyond isolated software tools and becoming integrated into the systems, institutions, and infrastructure that support modern society. As these systems grow more interconnected, influential, and capable, questions of trust become increasingly important. The challenge is no longer limited to what intelligent systems can do. It extends to the conditions under which they operate, and the degree to which people can meaningfully participate in, understand, influence, and benefit from them.

Historically, major advances in human capability have often preceded the constitutional, institutional, and social conditions needed to govern them responsibly. The Industrial Revolution demonstrated how transformative capability can outpace the structures needed to guide it. The Digital Age presents a similar challenge: AI infrastructure is becoming embedded in public life before trust architecture is fully established.

This paper proposes Trust Architecture as a constitutional framework for AI infrastructure. The framework begins with a simple but consequential observation: trust develops through conditions rather than through imposition, declaration, or after-the-fact additions. Those conditions determine whether participation is meaningful, whether institutions are viewed as legitimate, whether accountability can be maintained, and whether increasingly intelligent systems remain aligned with human responsibility.

Within this framework, *constitutional* refers to the foundational conditions that shape how legitimacy, participation, accountability, limits, responsibilities, correction, and continuity are organized within AI infrastructure. The term does not propose a single legal code, state constitution, or regulatory regime. It refers instead to the deeper structural conditions through which trustworthy systems can be built, evaluated, corrected, and renewed.

Governance frameworks can define rules, institutions, safeguards, and enforcement mechanisms. Trust Architecture addresses the constitutional conditions those frameworks must preserve if AI infrastructure is to support genuine trust, legitimate participation, accountability, correction, and stewardship over time.

The framework is organized around the concept of Genuine Trust and Legitimate Participation. The section that follows — The Conditions of Trust — introduces this concept and provides the architectural foundation for the seven constitutional conditions examined throughout: Meaningful Participation, Human Dignity and Sovereignty, Privacy, Trustworthy Representation, Institutional Integrity and Accountability, Correction and Renewal, and Stewardship Across Generations.

These conditions are examined individually for clarity, but their significance emerges most clearly through their relationships to one another. They lead, finally, into a synthesis: Human Continuity and Collective Advancement — a consideration of why the constitutional conditions matter together, and how they preserve the conditions of a civilized world in which humanity can continue to participate, learn, create, discover, adapt, and responsibly extend its capabilities.

The purpose of this paper is to clarify the constitutional conditions that support trustworthy AI infrastructure, and to contribute to the ongoing development of frameworks capable of sustaining genuine trust, legitimate participation, accountability, stewardship, and human continuity in an increasingly interconnected world.

Genuine Trust and Legitimate Participation — The Constitutional Objective

The constitutional objective of this framework is Genuine Trust and Legitimate Participation. These two concepts are distinct but inseparable within trustworthy AI infrastructure.

Trust without participation risks becoming passive reliance. Participation without trust risks becoming symbolic, extractive, or hollow. Together, they describe the conditions under which people can engage with systems, institutions, and technologies in ways that are meaningful, accountable, and worthy of reliance.

Genuine trust differs from the appearance of trust. Branding, claims, authority, or convenience alone cannot produce it. Genuine trust develops when systems demonstrate integrity, accountability, reliability, reciprocity, correction, and continued connection to reality. It is strengthened when people can observe that the systems affecting them are answerable, understandable, and capable of being challenged and improved.

Legitimate participation is more than access or formal inclusion. It requires conditions under which people can understand, question, influence, contribute to, and seek correction within systems that affect their lives. Participation becomes legitimate when surrounding structures support dignity, privacy, accountability, representation, and fairness. When these conditions are absent, participation can become an act of compliance rather than a meaningful civic or human role.

The objective of Genuine Trust and Legitimate Participation gives the framework its constitutional direction. The sections that follow examine the conditions that make this objective more than an aspiration — describing how trust develops, how participation becomes meaningful, how human dignity and sovereignty are protected, how privacy supports trust, how representation keeps the framework connected to reality, how institutions remain accountable, how correction and renewal keep systems honest, and how stewardship preserves continuity across generations.

The Conditions of Trust

Trust is not a mechanism, feature, institutional label, or claim made by authority. Trust develops through conditions that allow people to rely upon systems, institutions, and relationships with confidence. Those conditions include accountability, reciprocity, transparency, privacy, dignity, representation, correction, stewardship, and demonstrated responsibility.

Trust is often most visible through its absence. When trust weakens, participation becomes guarded, cooperation becomes costly, institutions lose legitimacy, and societies compensate through verification, enforcement, surveillance, and control. These mechanisms can reduce immediate risk, but they only partially replace trust itself.

Security remains necessary in any complex society. The concern is the pursuit of security without sufficient trust, legitimacy, accountability, privacy, participation, and stewardship. Contemporary conflicts between nations illustrate how security pursued in isolation can produce instability, suspicion, and recurring cycles of escalation. When individuals, institutions, or nations seek security through control, secrecy, surveillance, or force alone, these measures can reduce immediate risk while deepening the very conditions they seek to manage.

Trust Architecture assumes that trust will be tested. In relationships among institutions, nations, and systems, trust can be probed, imitated, challenged, or exploited before it becomes sufficiently demonstrated and reinforced. Mature trust therefore requires continued awareness, verification, accountability, correction, and demonstrated conduct over time.

The pace of the Digital Age creates increasing demand for efficient, meaningful, and trustworthy participation. Institutions, nations, and systems have reason to participate in Trust Architecture because durable participation becomes difficult where trust remains weak, counterfeit, unstable, or absent. Trust Architecture supports legitimacy, accountability, cooperation, and long-term stability within increasingly interconnected environments.

Trust Architecture recognizes a world with risk, conflict, competition, crime, deception, and bad actors. It seeks constitutional conditions through which security and trust can develop together rather than in opposition. Trust changes how risk is understood, managed, and corrected.

As AI infrastructure becomes increasingly integrated into communication, education, commerce, governance, identity, finance, and public life, trust extends beyond technical capability. AI infrastructure must be worthy of trust rather than merely capable of claiming it. The constitutional conditions that follow describe the recurring supports through which trust is established, maintained, tested, corrected, renewed, and carried forward.

Meaningful Participation

Governance systems and AI infrastructure must protect and enable informed, inclusive, and effective participation — ensuring that individuals and communities have a meaningful voice in systems and decisions that affect them.

Meaningful participation is more than access, use, consent, or nominal representation. People can be required to use a system, provide information, or accept terms without truly participating in its direction, correction, or accountability. Participation becomes hollow when people are invited to provide input without any meaningful opportunity to explain, influence, question, or observe how that input is used. In such cases, participation becomes extractive: the system receives data while the participant receives little recognition, agency, or reciprocal value.

Meaningful participation is an active civic force. It allows the public to press for broader inclusion, greater access, fair treatment, accountable compliance, consumer protection, and avenues for redress. It helps move systems toward legitimacy by insisting that they remain open, fair, responsive, and answerable to the people they affect.

As AI infrastructure becomes increasingly embedded in communication, education, commerce, governance, and public life, participation must remain connected to voice, agency, transparency, informed understanding, due process, and meaningful opportunities to influence, challenge, or seek correction.

Where meaningful participation weakens, decision-making authority becomes disconnected from those subject to its consequences. This creates risks to legitimacy, accountability, trust, and democratic governance — and signals the possible concentration of influence, information, or control within particular institutions or interests.

Trust Architecture therefore identifies the legal, institutional, and technical conditions that keep participation informed, protected, reciprocal, and capable of shaping the infrastructure that increasingly shapes society.

Human Dignity and Sovereignty

Human dignity recognizes the inherent value, standing, and worth of persons. Human sovereignty protects the ability of individuals to exercise agency, consent, identity, self-direction, and meaningful participation in matters that affect their lives.

Dignity and sovereignty are closely related. Dignity concerns how human beings are recognized and treated. Sovereignty concerns the practical conditions that allow individuals to exercise autonomy, judgment, and choice. Together they preserve the human foundations upon which trust and legitimate participation depend.

Rights, protections, and due process are necessary conditions for dignity and sovereignty to exist in practice. When these protections are absent, agency weakens, consent becomes questionable, and participation becomes symbolic rather than meaningful.

Human dignity is experienced by individuals but protected through collective responsibility. Societies, institutions, technologies, and governance systems shape the conditions under which dignity and sovereignty flourish or deteriorate. These systems are composed of people, affect people, and exist in relationship to people — a fact that any serious observation of the whole must recognize.

As AI infrastructure becomes increasingly integrated into communication, education, commerce, governance, identity, and public life, preserving dignity and sovereignty becomes increasingly important. Systems that influence access, opportunity, reputation, information, or decision-making must remain accountable to the people they affect.

Trustworthy systems distinguish genuine empowerment from its appearance. Systems that claim to increase participation while reducing agency, consent, autonomy, or meaningful choice risk producing counterfeit trust. The appearance of dignity and sovereignty is a poor substitute for their genuine protection.

Dignity and sovereignty also protect individuals from manipulation, coercion, exploitation, predation, and forms of participation that serve systems without adequately serving people. When dignity and sovereignty weaken, trust, legitimacy, and meaningful participation weaken with them.

Privacy

Privacy protects human dignity, sovereignty, participation, autonomy, security, and trust. It preserves the ability of individuals to think, communicate, learn, and participate without unnecessary exposure, surveillance, manipulation, coercion, or exploitation.

Privacy is not absolute secrecy. Different roles, responsibilities, and institutions carry different expectations of transparency and accountability. Individuals generally require strong privacy protections, while institutions exercising public authority require greater transparency to support legitimacy and accountability.

Privacy is a foundational condition for trust. Trust struggles to solidify where privacy is routinely compromised. When individuals repeatedly experience unauthorized access, data breaches, misuse of personal information, identity theft, or loss of control over their data, confidence in digital participation weakens — not only in particular institutions, but in the broader digital systems people are increasingly asked to rely upon.

Privacy includes meaningful control over personal information. Certain categories of personal data can reasonably remain outside centralized, continuously shared, or permanently stored repositories when local or individual control is practical. Individuals retain the ability to maintain, manage, and selectively share personal information through personal devices, encrypted systems, trusted intermediaries, or other mechanisms that preserve agency, consent, and sovereignty.

Centralized storage of personal data creates concentrated vulnerability. When institutions hold large amounts of personal information, failures of security, stewardship, or accountability can expose individuals to harm beyond their control. Trust Architecture therefore encourages continued examination of privacy-preserving approaches: responsible stewardship, data minimization, encryption, selective sharing, and other methods that strengthen trust while supporting legitimate participation.

The gradual normalization of privacy loss can create the perception that individuals have little meaningful control over their personal information. Trust Architecture challenges this assumption. Future generations can reasonably expect stronger forms of privacy, agency, stewardship, and control than those commonly accepted today.

Privacy therefore extends beyond a present concern. It is also an act of stewardship — protecting the conditions under which future participants continue to exercise dignity, sovereignty, agency, and trust within increasingly intelligent and interconnected systems.

Trustworthy Representation

Trustworthy Representation ensures that the realities necessary for genuine trust and legitimate participation remain visible within the framework.

Within this constitutional framework, representation extends beyond elected representatives, political advocacy, or formal constituency seats. It refers to the reliable visibility of affected people, lived experience, expertise, risks, consequences, future concerns, emerging realities, and changing conditions within observation, governance, certification, correction, and decision-making.

Reality is dynamic rather than fixed. New conditions, capabilities, risks, opportunities, and forms of understanding continue to emerge. Trustworthy Representation keeps the framework connected to these evolving realities through continual observation, reciprocal learning, challenge, correction, and adaptation.

Trustworthy Representation is the oxygen of the framework. It continually introduces new observation, consequence, experience, expertise, and corrective challenge into the system. When this oxygen is absent, stagnation follows — systems continue operating according to assumptions or priorities that no longer accurately reflect the realities they are meant to serve.

Observation alone is insufficient. Observation must remain connected to the realities it seeks to understand and serve. Representation becomes untrustworthy when observation grows disconnected from reality, when assumptions replace evidence, or when authority becomes insulated from challenge and correction.

No individual, institution, discipline, technology, community, or generation observes the whole. Each perspective carries strengths, limitations, assumptions, and blind spots. Trustworthy Representation therefore requires structured pathways for reciprocal learning, challenge, correction, and broader observation.

Trustworthy Representation avoids creating a fixed class of approved observers. Reality continues to evolve, and no observer, institution, or generation can claim permanent authority over its interpretation. As human and technological capabilities develop, new forms of observation, analysis, and learning emerge. The framework preserves conditions through which new observations, expertise, perspectives, and corrective insights can emerge as circumstances change — while ensuring that all forms of observation remain subject to accountability, challenge, and reciprocal learning.

Through reciprocal learning and correction, Trustworthy Representation helps prevent the framework from becoming sealed inside its own assumptions — ensuring that reality remains capable of continually informing the system.

Institutional Integrity and Accountability

Institutions, organizations, systems, and authorities that seek public trust must themselves operate within Trust Architecture. Trust requires reciprocal standards of integrity, accountability, fairness, transparency, correction, and responsible stewardship.

Organizations have reason to participate in Trust Architecture when doing so strengthens legitimacy, public confidence, regulatory readiness, institutional resilience, and durable participation. Participation is not assumed. It becomes more likely when trust-supporting conditions align with organizational interest, public expectation, market confidence, and governance requirements.

Institutional integrity maintains alignment between claims and reality. Representations, certifications, decisions, communications, and public commitments must remain connected to observable facts, evidence, and verifiable conditions. Integrity protects against deception, misrepresentation, counterfeit trust, and the erosion of confidence that occurs when claims become disconnected from reality.

Institutional accountability maintains alignment between actions and responsibility. Decisions, outcomes, and exercises of authority must remain attributable, examinable, challengeable, correctable, and subject to consequence where appropriate. Accountability helps ensure that power remains answerable to those affected by it.

In governmental contexts, checks and balances can operate through formal constitutional structures. In private, institutional, or technical contexts, Trust Architecture depends more often on accountability, transparency, independent review, certification, correction mechanisms, public scrutiny, and regulatory oversight where applicable.

Trust Architecture requires reciprocity. Individuals are often asked to provide information, comply with requirements, and place trust in institutions. Institutions that request such trust must remain equally accountable to the people they affect. When reciprocity is absent, relationships become skewed — concentrating visibility, information, and power in one direction while reducing trust, participation, and legitimacy in the other.

Institutional fairness requires attention to the broader whole. Systems, standards, certifications, and governance processes lose integrity when shaped primarily by the most influential, visible, or well-resourced interests. Institutions must remain attentive to constituencies that are less visible, less organized, underrepresented, or unable to advocate effectively for themselves.

Certification can provide important trust scaffolding when it reflects demonstrated integrity, accountability, safety, fairness, and responsible development. An institution or system that lacks certified trustworthiness has no legitimate claim to a certified label. Certification represents a higher demonstrated standard, not merely a procedural one.

Certification is a trust path, not a trust gate. It recognizes demonstrated trust development without becoming a monopoly on legitimacy or a barrier to responsible participation, innovation, and future advancement. Voting offers a useful illustration: participation becomes legitimacy-bearing because the process is governed by recognized standards, verification, and accountability. The broader principle is that certification makes consequential participation more trustworthy rather than merely more controlled.

Institutional integrity and accountability strengthen genuine trust and legitimate participation by ensuring that authority remains connected to reality, responsibility, fairness, reciprocity, and public examination. Legitimacy emerges through continued trustworthy conduct — not through declaration alone.

Correction and Renewal

Correction and Renewal recognizes that no system, institution, framework, or constitutional condition is complete at its inception. Trustworthy AI infrastructure must remain open to being examined, challenged, corrected, and improved — not as a sign of weakness, but as a condition of integrity.

Correction is the mechanism through which errors, failures, misalignments, and harms are identified and addressed. Without structured pathways for correction, systems accumulate errors over time. Institutions become resistant to challenge. Frameworks harden around assumptions that no longer reflect reality. The absence of correction does not indicate the absence of problems — it indicates the absence of the conditions needed to surface and address them.

Renewal extends beyond correction. Where correction addresses what has gone wrong, renewal asks what must be actively cultivated to remain trustworthy over time. Systems, institutions, and constitutional frameworks require ongoing investment, examination, and adaptation to remain legitimate and connected to the realities they serve. Renewal is the generative counterpart to correction — together they sustain the capacity for continued improvement.

Correction and Renewal are conditions for all other constitutional conditions. Meaningful participation requires correction when voices are excluded or ignored. Human dignity requires renewal when protections erode. Privacy requires correction when data practices drift from their stated purposes. Trustworthy Representation requires renewal when observation grows stale or disconnected. Institutional integrity requires correction when claims and conduct diverge. Stewardship requires renewal when inherited frameworks no longer serve the realities of the present.

The iterative development of this framework illustrates Correction and Renewal in practice. Each version examines what was previously stated, identifies what was omitted or imprecise, and works to improve the whole. That process is not incidental to the framework — it is an expression of its constitutional purpose. Trustworthy systems are not those that claim perfection. They are those that remain genuinely open to examination, challenge, and improvement.

Correction and Renewal therefore require more than good intentions. They require structured conditions: accessible channels for challenge and redress, institutional willingness to examine and revise, transparency sufficient to identify where correction is needed, and cultures that treat improvement as a sign of strength rather than an admission of failure.

In an era of rapidly advancing AI infrastructure, the capacity for correction and renewal is not a luxury. It is a constitutional necessity. Systems that cannot be corrected cannot be trusted. Frameworks that cannot be renewed cannot endure.

Stewardship Across Generations

Stewardship Across Generations seeks to preserve and renew the conditions necessary for trustworthy civilized life across time.

Humanity inherits knowledge, institutions, infrastructure, culture, rights, responsibilities, achievements, and consequences from previous generations. Each generation receives a world shaped in part by those who came before, and contributes to the conditions inherited by those who follow.

The purpose of stewardship extends beyond preservation or change in isolation. It is the responsible care, improvement, and transmission of constitutional conditions that support genuine trust, legitimate participation, human dignity, privacy, accountability, correction, and continuity.

The Industrial Revolution illustrates what happens when transformative infrastructure advances without sufficient constitutional guidance. Labor protections, safety standards, and regulatory structures often emerged after exploitation and unsafe conditions had already become deeply embedded. The lesson is clear: transformative capability requires trust structures early enough to guide participation, protection, and correction.

The Digital Age is unfolding under similar conditions of rapid capability, connectivity, automation, dependency, and institutional strain. Trust Architecture is proposed accordingly — as a constitutional framework for AI infrastructure designed to protect genuine trust, legitimate participation, dignity, privacy, accountability, and continuity before harmful patterns harden into infrastructure.

Stewardship operates without predicting the future. History demonstrates that each generation encounters new challenges, opportunities, responsibilities, and capabilities that were not fully anticipated. While future conditions remain uncertain, experience suggests that future generations will inherit both opportunities and challenges beyond those of the present.

Stewardship passes forward constitutional principles along with the ability to examine, challenge, refine, and improve them responsibly. Preservation without adaptation becomes rigidity. Adaptation without principles becomes instability. Stewardship preserves continuity while remaining open to learning, correction, and renewal.

Constitutional principles endure when successive generations continue to find them valuable, legitimate, and worthy of care. Trust Architecture is offered in the hope of constitutional trust development: the continued cultivation of conditions that support a civilized, participatory, and trustworthy society.

The framework leaves future values, future technologies, and humanity's direction open — while preserving the conditions through which future generations continue to participate, learn, create, discover, achieve, and responsibly extend human capability.

Stewardship Across Generations helps ensure that Trust Architecture remains capable of continuity, renewal, responsibility, and constructive evolution. Through stewardship, humanity preserves the conditions that allow both continuity and advancement to remain possible.

Human Continuity and Collective Advancement — Synthesis

The seven constitutional conditions explored throughout this framework are examined individually for clarity. Their broader significance, however, emerges through their relationships to one another.

Viewed collectively, these conditions form Trust Architecture — supporting human continuity and collective advancement. They help establish the conditions through which people, institutions, and increasingly intelligent systems participate within a shared environment sustained by legitimacy, accountability, responsibility, and continued connection to reality.

Human continuity extends beyond preservation alone. A society that preserves itself without learning, adapting, or expanding opportunity risks stagnation. Collective advancement extends beyond capability alone. Advancement that outpaces legitimacy, accountability, participation, dignity, privacy, correction, or stewardship creates instability and undermines the very conditions that support long-term progress.

History illustrates this tension. The Industrial Revolution produced extraordinary advances in production, transportation, communication, and economic capacity. Many of the constitutional, institutional, and social conditions needed to govern those capabilities responsibly emerged later — through adaptation, correction, negotiation, and continued development. The Digital Age presents the same challenge. As intelligent systems become increasingly integrated into public life, questions of trust, participation, accountability, privacy, representation, correction, and stewardship become inseparable from questions of capability itself.

Trust Architecture addresses this challenge at the level of infrastructure. Trust is treated as part of the constitutional architecture within which systems operate — not as an external expectation applied after systems are built. The seven constitutional conditions work as an architectural foundation supporting legitimacy, accountability, participation, and human responsibility within AI infrastructure.

The continuity of a civilized world depends upon preserving constitutional principles and advancing them in ways that remain relevant to changing realities. Adoption, agreement, and success remain contested. Constitutional frameworks provide structures through which societies observe, evaluate, challenge, refine, and improve the conditions under which participation occurs.

Human continuity and collective advancement serve as the synthesis of the framework — reconnecting the constitutional conditions to their broader purpose: preserving and advancing the conditions through which humanity continues participating in shaping its own future while responsibly extending its capabilities. In this sense, Trust Architecture is part of an ongoing process of constitutional development, stewardship, correction, and renewal.

Conclusion

Artificial intelligence is becoming part of the infrastructure through which people communicate, learn, work, govern, participate, and make decisions. As this integration deepens, the central question is no longer only what intelligent systems can do. The deeper constitutional question is what conditions shape their operation — and whether those conditions support trust, dignity, accountability, participation, stewardship, and human continuity.

History shows that capability can advance faster than the conditions needed to guide it responsibly. The Industrial Revolution demonstrated this pattern clearly. The Digital Age is repeating it: AI infrastructure is becoming embedded in public life before trust architecture is fully established. Technological capability continues to accelerate, but capability alone cannot create legitimacy, participation, trust, or a more civilized world.

Trust Architecture responds by treating trust as a structural condition of AI infrastructure rather than an external expectation. Trust develops through seven constitutional conditions that make participation meaningful, dignity protected, privacy respected, representation connected to reality, institutions accountable, correction possible, and stewardship sustained across generations.

These conditions form an architecture because their strength emerges through relationship. Trust without accountability weakens. Participation without dignity becomes shallow. Privacy without stewardship becomes fragile. Representation without correction becomes stagnant. Advancement without continuity becomes unstable. Together, the seven constitutional conditions support the possibility of genuine trust and legitimate participation within AI infrastructure.

This framework provides architectural clarification rather than an engineering specification or final design. It clarifies constitutional conditions that future engineering, governance, certification, and institutional development can build upon. Version 3 therefore remains architectural in purpose: it makes visible the structure of trust that future development must continue to test, refine, and build upon.

The future remains open. New capabilities will emerge, new risks will appear, and future generations will inherit conditions they did not create. Trust Architecture preserves and advances the conditions through which people continue participating in the creation of that future.

The architecture is now visible. Its continued development points toward future constitutional trust work: a more formal expression of the principles, trust conditions, and engineering responsibilities required for trustworthy AI infrastructure. Digital architecture can be modified, corrected, and renewed. Its continued development belongs to those who build, govern, steward, challenge, improve, and inherit what follows.